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Lab Procedure
Infrared Distance

Introduction

Ensure the following:

1. You have reviewed the **Application Guide – Infrared Distance**
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
    a. The LED next to the USB C port will stay white after the touch screen starts loading.
    b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

Exploring Infrared Distance Data

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder**..., and browse to the directory containing the python files and lab procedure for the Encoders lab (.../sensors_trainer/1_fundamentals /encoder/hardware/python). Open this directory.
2. We will be recording data using the infrared Distance sensor. Open **GP2Y0A21YK0F_Spec_Sheet.pdf** and identify the minimum and maximum measurement range.
3. Draw a distance ruler from the front of the Mechatronic Sensor Trainer start from 0cm to the maximum range.
4. Grab a flat surface and place it so the flat surface is parallel to the front of the mechatronic sensor board.
5. Run **infrared_display.py** using the ▷ button or the terminal. Slowly move the flat surface close to the Mechatronics trainer and further away.
6. How does the voltage measurement look like?
7. Open **infrared_record.py**. This experiment will require you to record voltage measurements used to generate the following equation:

$$d = Ax^b \tag{1}$$

8. You will need to define the following parameters:

    a. `minMeasurementValue`

    b. `stepBetweenMeasurements`

    c. `maximumMeasurementValue`

    Since we are using sampling methods to identify the value of A and b, it's important to generate as many samples as possible. We are also assuming the distance value being calculated is in meters.

9. Place a flat object at marker for minimum distance. It's recommended to keep the flat object parallel to the front surface of the Mechatronic Trainer sensor.

10. Run **infrared_record.py** using the ▷ button or the terminal. Click on the terminal session and use the spacebar on your computer to record a sample.

11. Move the object based on the value specified for `stepBetweenMeasurements`.

12. Once the total number of samples is reached a line fitting technique will be used to estimate A and b and a plot will show the predicted model of the infrared distance sensor compared to the actual sampled value.

13. Do the calibrated A and b values match the distance/voltage measurements? If so, make note of the A and b v values.

14. Open **infrared_record.py** and change:

    a. `amplitude` to the calculated value of A.

    b. `exponent` to the calculated value of b.

15. Run **infrared_record.py** using the ▷ button or the terminal.

16. A plot will show you both the Voltage and Distance measurements from the Infrared Distance sensor. Keep in mind the Distance is assumed to be in meters.

17. Once again use the distance ruler to verify the distance measurement reported.

18. How well does the inferred model in steps 7-8 match the actual physical distance?

19. Stop, save and close your program.

20. Open **infrared_distance.py** and change:

    a. amplitude for the recorded value of A

    b. exponent for the recorded value of b

21. Run **infrared_ distance.py** using the ▷ button or the terminal.

22. Using the distance ruler, move the infrared sensor to the distances used for calibration and make a note of the displayed distance values. Do they match correctly? What other considerations might you need to take when using an infrared distance sensor?

23. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.